Sprint Backlog and Plan

Software Engineering - SVNC.00.308

Project: StayEase Hotel Reservation System

Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustiuhov)

Date: March 2, 2026

Version: 1.0

Project Name: StayEase Hotel Reservation System
Team: Cat Bacillus (Kyrylo Pryiomyshev, Danylo Chernov, Daniil Ustiuhov)
Sprint Number: 1
Sprint Duration: 90 minutes (Lecture 5 + Lecture 6)
Date: March 2, 2026

Sprint Goal

Goal: Enable guests to search for available rooms, register an account, and make a reservation.

Capacity Planning

Estimated Capacity: 21 story points (5 user stories)

Rationale: This is our first sprint. Without historical velocity data, we estimated based on 90 minutes of implementation time, a 3-person team, moderate complexity of web-based reservation logic, and a buffer for unknowns. Each team member can reasonably deliver 7 story points in 90 minutes.

Team Availability: All 3 members available for full sprint.

Selected User Stories

Story ID	User Story Summary	Priority	Estimate	Status
US-01	Search available rooms by dates and guest count	Must-have	M (5)	To Do
US-02	View detailed room information	Must-have	S (3)	To Do
US-03	Guest registration and login	Must-have	M (5)	To Do
US-04	Make a reservation for selected room and dates	Must-have	M (5)	To Do
US-05	Prevent double booking on overlapping dates	Must-have	S (3)	To Do

Total Committed: 5 stories, 21 story points

Task Breakdown

Story	Task Description	Owner	Est. Time	Status
US-01	Design room search API endpoint with date range query	Kyrylo	20 min	Done
US-01	Implement search results UI with room cards	Danylo	20 min	Done
US-01	Test search with various date ranges and edge cases	Daniil	15 min	Done

US-02	Implement room detail page with amenities and photos	Danylo	15 min	Done
US-02	Connect room detail to search results navigation	Danylo	10 min	Done
US-03	Implement registration form with validation	Danylo	15 min	Done
US-03	Implement login API with JWT token generation	Kyrylo	20 min	Done
US-03	Test registration and login flows	Daniil	10 min	Done
US-04	Implement reservation creation API with availability check	Kyrylo	20 min	Done
US-04	Build booking confirmation UI with summary	Danylo	15 min	Done
US-04	Test booking flow end-to-end	Daniil	15 min	Done
US-05	Implement date overlap detection query	Kyrylo	15 min	In Progress
US-05	Add concurrent booking guard with DB transaction	Kyrylo	15 min	Not Started
US-05	Test double booking scenarios	Daniil	10 min	Not Started

Sprint Progress Tracking

Day 1 (Lecture 5)

Completed

- US-01: Room search API and UI fully implemented and tested
- US-02: Room detail page implemented and connected to search
- US-03: Registration form started (validation in progress)

In Progress

- US-03: Login API implementation (JWT generation)

Blockers

- None

Day 2 (Lecture 6)

Completed

- US-03: Registration and login fully implemented and tested
- US-04: Reservation creation API, confirmation UI, and end-to-end tests done

In Progress

- US-05: Date overlap detection query written but not fully tested

Blockers

- US-05: Concurrency handling for simultaneous booking attempts turned out more complex than estimated. Ran out of time before implementing transaction-level locking.

Sprint Notes

Decisions

- Decided to use PostgreSQL date range operators (OVERLAPS) for availability checking
- Agreed to use JWT with 24-hour expiry for session management
- Chose to store prices as DECIMAL(10,2) to avoid floating-point issues

Assumptions

- Check-in time is 14:00 and check-out time is 12:00 (standard hotel policy)
- Room prices are fixed and do not vary by date or season in V1

Technical Notes

- PostgreSQL OVERLAPS operator simplified date range queries significantly
- bcrypt hashing adds ~250ms per registration — acceptable for current load

Blockers and Resolutions

Blocker: Concurrent booking prevention requires database-level transaction isolation

Resolution: Deferred to Sprint 2. Basic overlap check works for sequential bookings; transaction-level locking will be added as part of US-05 completion.

Sprint Commitment

Kyrylo Pryiomyshev: ________________ Date: March 2, 2026

Danylo Chernov: ________________ Date: March 2, 2026

Daniil Ustiuhov: ________________ Date: March 2, 2026